1 **(a)** displacement is a vector, distance is a scalar — B1
displacement is straight line between two points / distance is sum of lengths moved / example showing difference — B1 [2]
(either one of the definitions for the second mark)

(b) a body continues at rest or at constant velocity unless acted on by a <u>resultant</u> (external) force — B1 [1]

(c) **(i)** sum of T_1 and T_2 equals frictional force — B1
these two forces are in opposite directions — B1 [2]
(allow for 1/2 for travelling in straight line hence no rotation / no resultant torque)

(ii) **1.** scale vector triangle with correct orientation / vector triangle with correct orientation both with arrows — B1
scale given or mathematical analysis for tensions — B1 [2]

2. $T_1 = 10.1 \times 10^3$ $(\pm 0.5 \times 10^3)$ N — A1
$T_2 = 16.4 \times 10^3$ $(\pm 0.5 \times 10^3)$ N — A1 [2]